



Assignment of bachelor's thesis

Title: Spatiotemporal interpolation of precipitation and machine learning
Student: Tamerlan Ismakatov
Supervisor: Ing. Ondřej Podsztavek, Ph.D.
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Instructions

Climate projections (i.e. climate simulations typically until 2100) with high spatial resolution are necessary for local decision-making, such as urban planning or forest management. However, only climate projections with low spatial resolution are usually available, which are often insufficient. Therefore, increasing their spatial resolution (i.e. downscaling) is important. However, downscaling needs high-resolution observational gridded data sets (e.g. E-OBS, Nordic Gridded Climate Dataset). These data sets are created from weather station measurements.

Creating such data sets can be approached with machine learning methods such as gradient boosting (e.g. XGBoost, LightGBM, CatBoost) and Bayesian neural fields. Besides weather station measurements, these methods can incorporate auxiliary data such as elevation (e.g. Copernicus DEM), land use (e.g. SoilGrids), etc.

The goal is to create an observational gridded data set for precipitation in resolutions of 1 km or less using machine learning.

1. Review literature on spatiotemporal interpolation using machine learning.
2. Implement a baseline method (e.g. kriging) and at least two state-of-the-art probabilistic machine learning methods.
3. Experiment with the methods on the data set from Regionales Klimainformationssystem (ReKIS) combined with auxiliary data (e.g. elevation, land use).



4. Evaluate the methods with relevant metrics (e.g. mean absolute error, continuous ranked probability score).
5. Use the best method to create observational gridded data sets in different spatial resolutions.
6. Discuss the results concerning different spatial resolutions and the importance of auxiliary data.

Bibliography

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